

Building a Kinetic Mobility Database for the Fe-Ni-Nb System

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Abstract

This study attempts to build a kinetic mobility database for the iron-nickel-niobium (Fe-Ni-Nb) system. This system was selected as a ternary representation of Inconel 718 (IN718), which is a nickel-base alloy that can attribute its high temperature performance to the precipitation of γ'' and δ , two crystal structures of Ni_3Nb . The study begins with the development of a Fe-Ni-Nb ternary thermodynamic database (.tdb) compatible with Pandat 2025. After demonstrating the functionality of this thermodynamic database, mobility models are added to the thermodynamic database to extend the database's functionality from prediction to equilibrium phases to the modeling of kinetic processes, such as diffusion. Finally, a kinetic mobility data (.kdb) is constructed to predict the precipitation of γ'' and δ phases in the Fe-Ni-Nb system. This study failed to predict the precipitation of these two strengthening phases in IN718, suggesting this ternary approximation of the alloy's composition is not a suitable approximation to model the critical phase transformations of IN718.

Keywords: Kinetic mobility database, Fe-Ni-Nb, CALPHAD, Inconel 718

I. Introduction

The modeling of materials is becoming increasingly important in the field of material science as a tool to accelerate novel alloy development. What was once done almost entirely empirically by metallurgists less than 100 years ago is now being pursued computationally, with the hopes of faster development cycles and more dramatic discoveries. This computational

approach to metallurgy is built on the CALPHAD (computation of phase diagrams) methodology and software tools, which requires thermodynamic and kinetic databases, known as .tdb and .kdb files respectively, to model materials. Therefore, the development of accurate databases based on experimental results and leading scientific theories is a necessary starting point for the modern material scientist.

This study attempts to write such databases for the Fe-Ni-Nb system. This system was selected as a ternary approximation of IN718, a nickel-based superalloy nominally composed of sixteen periodic elements [1]. The high element count of nickel-based superalloys makes them difficult to model due to the increase in complexity of atomic-scale interactions with each element introduced to the system. The ability to simplify these alloys into ternary approximations is of scientific value as it makes these alloys more manageable to model. However, these approximations can only be deemed appropriate if they capture the known, critical phase transformation of these alloys.

II. Methods

This study attempts to develop a kinetic mobility database for the Fe-Ni-Nb system following the CALPHAD approach [2]. The following section outlines the development of such a database, starting with the construction of a Fe-Ni-Nb thermodynamic database that can be used to calculate the ternary phase diagram for a given temperature. The thermodynamic database is then extended to include mobility data for the phases of interest in the ternary system considered. Finally, a kinetic mobility database is developed to predict the precipitation of IN718's critical secondary phases, γ'' and δ , to determine if the alloy can be approximated as a ternary system.

Throughout this study, Pandat 2025 was utilized as a CALPHAD software to calculate phase diagrams, concentration profiles, and precipitation using the thermodynamic and kinetic databases developed.

II.I Building a Fe-Ni-Nb Thermodynamic Database (.tdb)

The Fe-Ni-Nb system is a ternary that has been extensively studied due to its relevance to nickel-based superalloys. Therefore, experimental thermodynamic data necessary to construct a ternary database is readily available compared to alternative metallic systems, and evaluations of the binary databases included in the ternary database have occurred as recently as 2023 and 2024 [3, 4, 5]. Mathon et al. [6] presents a rich Fe-Ni-Nb thermodynamic database, which is used as the starting point for the .tdb developed in this study. The .tdb provided in Appendix B of [6] is formatted for Thermo-Calc, so it could not be opened in Pandat 2025 directly. The elements – Fe, Ni, and Nb – were not recognized, preventing the use of the database in Pandat. The syntax changes necessary to the .tdb to render the database functional in Pandat 2025 have been documented in the header of the .tdb [7].

To validate the successful translation of the .tdb from Thermo-Calc syntax to Pandat syntax, the binary subsystems of the Fe-Ni-Nb system were plotted along with ternary phase diagrams at three temperatures and compared to the phase diagrams established by Mathon et al [6]. With the thermodynamic database established for the system, mobility models were introduced to the .tdb to begin modeling the kinetic behavior of the system.

II.II Introducing Mobility Data to the Fe-Ni-Nb .tdb

The mobility framework adopted in this study is the multicomponent atomic-mobility formalism of Andersson and Ågren [8], which is the standard for CALPHAD-type kinetic databases and is implemented in Pandat 2025. In this formalism, the flux of species i relative to the lattice is expressed through an atomic mobility M_i . For a substitutional solution phase, the composition and temperature dependence of M_i is described using a Redlich–Kister polynomial form, which is also used for the Gibbs energy of each defined phase in the system. Following Andersson and Ågren [8], the product of the atomic mobility and the thermodynamic factor Γ_i is written as

$$RT \ln(M_i \cdot \Gamma_i) = \sum_p x_p \phi_i^p + \sum_p \sum_{q>p} x_p x_q \left[\sum_r \phi_i^{r,p,q} (x_p - x_q)^r \right] \quad (1)$$

where R is the gas constant, T is absolute temperature, and x_p is the mole fraction of component p . The first sum on the right-hand side contains the end-member parameters ϕ_i^p , which describe the mobility of species i in pure component p , and the second double sum contains the binary interaction parameters $\phi_i^{r,p,q}$, which capture deviations from ideal mixing of the kinetic behavior. Each end-member parameter is itself parameterized in Arrhenius form,

$$\phi_i^p = -Q_i + RT \ln D_0^i \quad (2)$$

with D_0^i the frequency factor and Q_i the activation energy for tracer diffusion of species i in pure p . For the scope of this study, all binary mobility interaction parameters $\phi_i^{r,p,q}$ were set to zero. This is equivalent to invoking a dilute-solution (or ideal-mixing) approximation for the kinetic description. The dilute-solution model approximates the mobility of each species in any composition using linear interpolation, weighted by mole fractions, between the corresponding end-member values. This simplification was made for two reasons: (1) it dramatically reduces the volume of experimental data required to add mobility data to the .tdb (composition-dependent interdiffusivity measurements are not required), and (2) the precipitation reactions critical to IN718, γ'' and δ , occur in a γ -Ni matrix (i.e. a Ni rich matrix). This means the ternary compositions of interests to evaluate whether Fe-Ni-Nb is a reasonable approximation of IN718 are near the Ni end-member, where the dilute-solution assumption may be valid. The trade-off is that the database is not expected to reproduce composition-dependent kinetic effects far from the Ni-rich corner of the ternary phase diagram.

To validate the integration of mobility data into the .tdb, the concentration profiles for binary and ternary diffusion couples were calculated using Pandat's PanDiffusion module. Table 1 outlines the diffusion couples simulated. The principal objective of calculating these concentration

profiles was to demonstrate that the mobility models introduced to the .tdb produced physically reasonable results before progressing to the construction of kinetic mobility database and precipitation related calculations for the ternary system. Note that terminal compositions were selected to ensure diffusion was only occurring in a FCC, γ -Ni matrix.

Table 1. Diffusion couples simulated

Run	l_{tot} [mm]	T [K]	t [hr]	C_L [at %]	C_R [at %]	Grid Size
1	1.0	1600	20	100Fe-0Ni-0Nb	0Fe-100Ni-0Nb	200
2	1.0	1200	20	100Fe-0Ni-0Nb	0Fe-100Ni-0Nb	200
3	1.0	1500	20	0Fe-95Ni-5Nb	0Fe-100Ni-0Nb	400
4	1.0	1473	20	20Fe-75Ni-5Nb	0Fe-100Ni-0Nb	400
5	1.0	1473	20	10Fe-83Ni-7Nb	0Fe-100Ni-0Nb	400

II.III Building a Fe-Ni-Nb Kinetic Mobility Database (.kdb)

To test the applicability of the Fe-Ni-Nb system as a ternary approximation of IN718, predictive power of precipitation of the alloy's critical secondary phases, γ'' and δ , is necessary. To do this, the thermodynamic database developed in Section II.I and II.II must be extended to a kinetic database, a .kdb, to include phase-specific kinetic inputs not contained in the .tdb. This section outlines the considerations made when developing a .kdb for the Fe-Ni-Nb system to target γ'' and δ precipitation, and how this database was evaluated via simulations using Pandat's PanPrecipitation module.

Pandat 2025 implements precipitation simulations through the PanPrecipitation module, which uses the Kampmann–Wagner Numerical (KWN) framework [9]. The particle size distribution of each precipitate phase is discretized into size bins, and concurrent nucleation, growth, and coarsening evolve in time using classical nucleation theory and a diffusion-limited growth law. The thermodynamic driving force and equilibrium tie-line compositions are obtained from the .tdb at every time step, and the diffusivities entering the growth law are obtained from

the mobility parameters appended to the .tdb in Section II.II. Phase-specific kinetic inputs not contained in the .tdb are read from the .kdb.

For each precipitate phase of interest, the .kdb specifies a molar volume, an interfacial energy σ between the precipitate and the matrix, an atomic spacing, a nucleation-site parameter that scales the available density of nucleation sites, and a driving-force factor. A separate parameter table is provided for the matrix phase (γ -Ni, which is named FCC_A1 in the databases) to define its molar volume. The interfacial energy and nucleation-site parameter are the two inputs to which the calculated volume fraction of precipitate is most sensitive, since together they control the nucleation barrier, Equation 3, and the pre-factor of the steady-state nucleation rate.

$$\Delta G^* \propto \frac{\sigma^3}{\Delta G_v^2} \quad (3)$$

Two precipitate phases were defined in the constructed .kdb, γ'' (named D022_NBNi3) and δ (named D0A_NBNi3). Both were modeled as spherical particles using the simplified, quasi-steady, diffusion-controlled growth law in PanPrecipitation, but were assigned different nucleation models, nucleation-site parameters, and interfacial energies to reflect the differences in these precipitates, Table 2. All .kdb parameters not discussed here were kept at their default values.

Table 2. Secondary phase .kdb kinetic inputs

Parameter	γ'' (D022_NBNi3)	δ (D0A_NBNi3)
Nucleation Model	Modified Homogeneous	Grain Boundary
Molar Volume [$\text{m}^3 \text{mol}^{-1}$]	7.5×10^{-6}	7.5×10^{-6}
Interfacial Energy [J m^{-2}]	0.025	0.300
Atomic Spacing [m]	3.60×10^{-10}	3.60×10^{-10}
Nucleation Site Parameter	0.01	0.001
Driving Force Factor	1	1

γ'' is a coherent intragranular precipitate, so it was assigned the modified-homogeneous nucleation model with a low interfacial energy of 0.025 J m^{-2} , consistent with the small lattice misfit between γ'' and the γ matrix. δ is incoherent and nucleates preferentially on grain boundaries, so it was assigned the grain-boundary nucleation model with a higher interfacial energy of 0.3 J m^{-2} . The nucleation-site parameter was set to 0.01 for γ'' and 0.001 for δ , reflecting the larger density of bulk sites for coherent intragranular nucleation relative to grain-boundary sites. Molar volumes ($7.5 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}$) and atomic spacings ($3.60 \times 10^{-10} \text{ m}$) were taken as equal for the two phases, consistent with their shared Ni_3Nb stoichiometry. The driving-force factor was left at unity for both phases.

Before any TTT or CCT diagrams were attempted, the .kdb was validated by single-temperature isothermal precipitation calculations on the loaded .tdb + .kdb combination. These checks serve two purposes: (1) confirming that the .kdb is parsed correctly and that both precipitate phases are recognized as nucleating, and (2) confirming that the predicted volume fractions evolve through nucleation, growth, and coarsening without reaching equilibrium instantaneously or remaining identically zero — either of which would indicate an input or coupling error rather than a meaningful kinetic prediction.

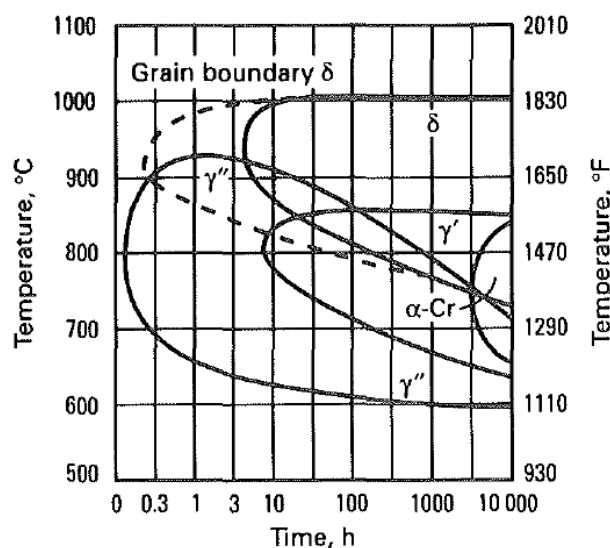


Figure 1. An established TTT diagram of commercial IN718 [10].

The benchmark composition was 22 at.% Fe, 3 at.% Nb, balance Ni, based on the limiting composition specifications for IN718 [1]. Isothermal holds were simulated at temperatures and times bracketing the “nose” of established TTT curves for γ'' and δ phases in commercial IN718, Figure 1 [10]. The volume fraction of each precipitate was calculated in Pandat using precipitation simulations in the PanEvolution/PanPrecipitation module.

III. Results

The following section presents the results of this study's efforts to produce a thermodynamic and kinetic database for the Fe-Ni-Nb system. Calculated results are compared to well-established results for the system, and commercial IN718 where applicable. A brief discussion accompanies the results in this section, followed by a more detailed discussion in Section IV.

III.1 Fe-Ni-Nb Phase Diagrams

Figure 2, Figure 3, and Figure 4 present the Fe-Ni, Ni-Nb, and Fe-Nb, binary phase diagrams, respectively, calculated using the modified .tdb developed during this study. For each of these figures, the binary phase diagram calculated by Pandat using the modified .tdb is presented on the left, while the equivalent phase diagram calculated by Mathon et al. [6] is presented on the right for comparison.

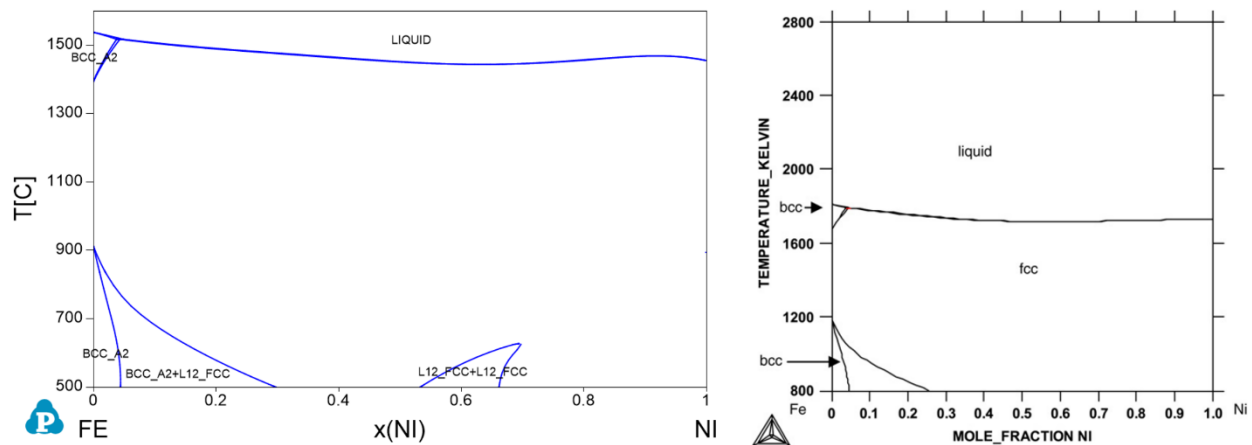


Figure 2. Fe-Ni phase diagram calculated by this study (left) and Mathon et al. [6] (right).

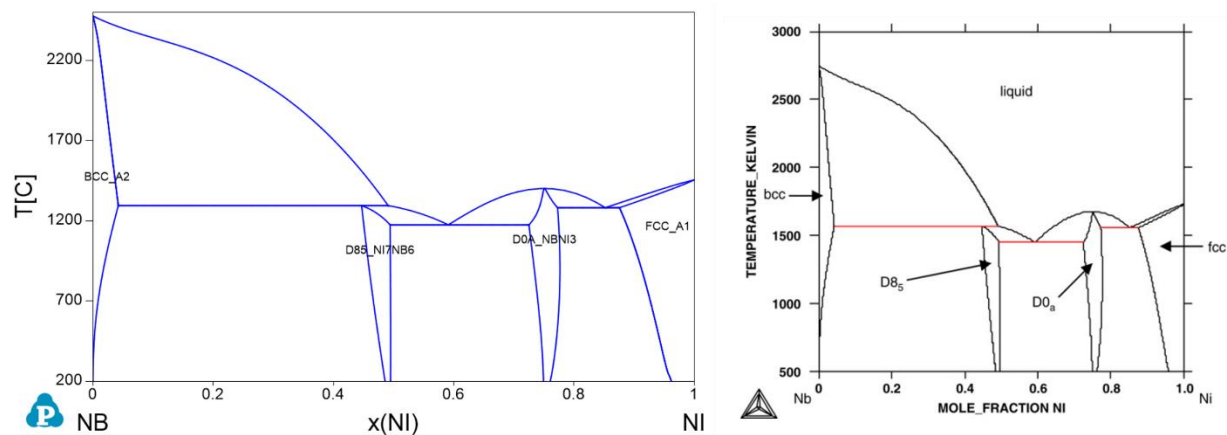


Figure 3. Ni-Nb phase diagram calculated by this study (left) and Mathon et al. [6] (right).

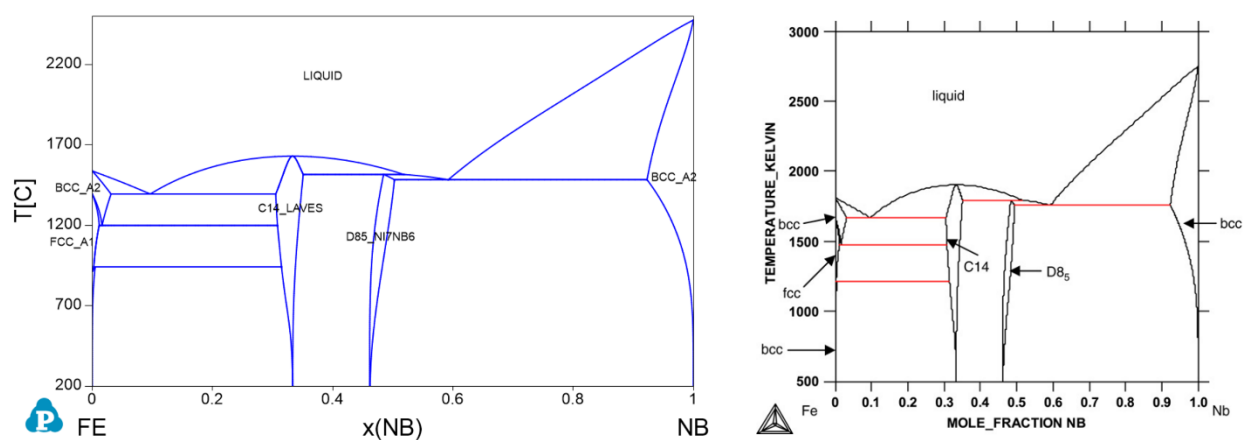


Figure 4. Fe-Nb phase diagram calculated by this study (left) and Mathon et al. [6] (right).

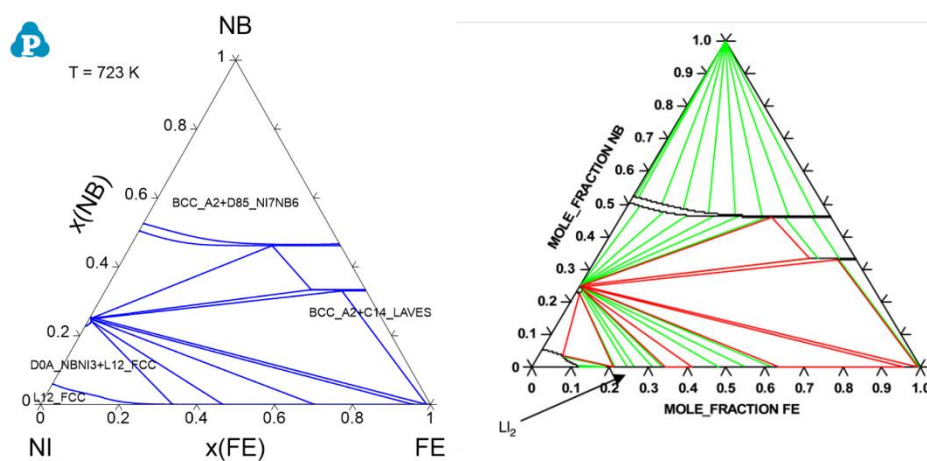


Figure 5. Ternary phase diagram calculated for $T = 723 \text{ K}$ by this study (left) and Mathon et al. [6] (right).

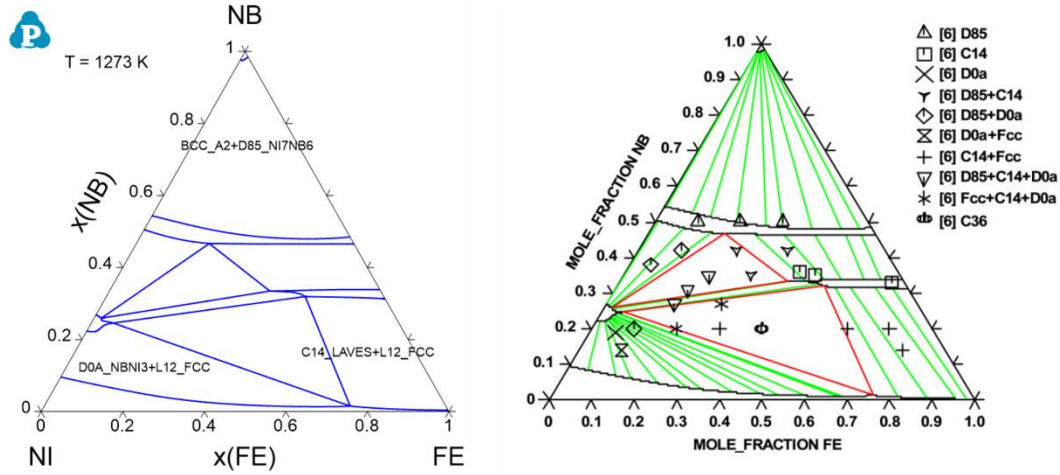


Figure 6. Ternary phase diagram calculated for $T = 1273$ K by this study (left) and Mathon et al. [6] (right).

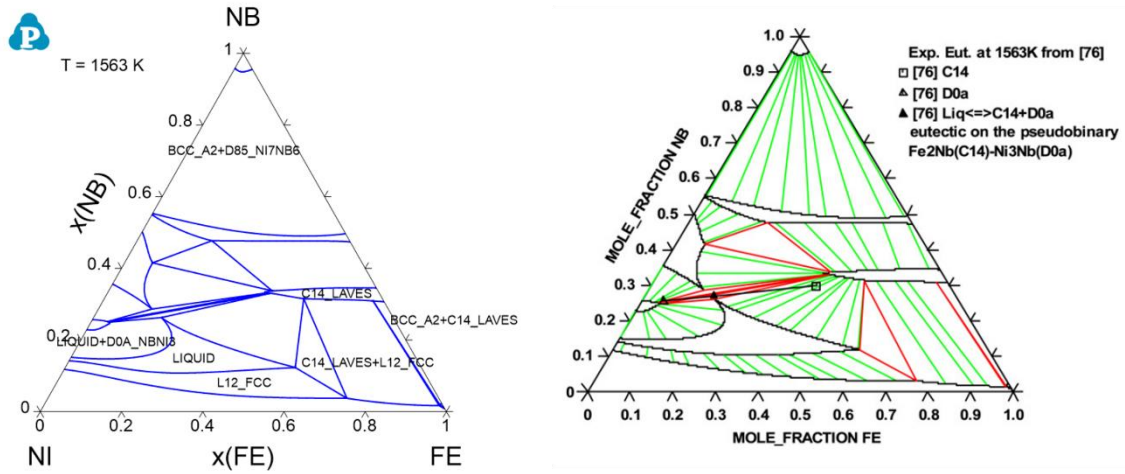


Figure 7. Ternary phase diagram calculated for $T = 1563$ K by this study (left) and Mathon et al. [6] (right).

Similar to the binary phase diagrams, isothermal ternary phase diagrams were calculated using the modified .tdb developed for this study, Figure 5, Figure 6, and Figure 7. Ternary phase diagrams were calculated at three different temperatures (723 K, 1273 K, and 1563 K) and compared to the ternary phase diagrams calculated by Mathon et al. [6] at these temperatures.

III.II Fe-Ni-Nb Diffusion Profiles

Table 3 presents the mobility data discussed in Section II.II. Note that tracer diffusivities for each species were only found for the FCC Ni and FCC Fe end-members, Equation 1. Tracer

diffusivities for each species of the FCC Nb end-member were assumed to be equivalent to the tracer diffusivities found for the FCC Ni end-member. This assumption was made because Nb exists as a BCC crystal structure for all temperatures up to its melting temperature, so tracer diffusivity in FCC Nb cannot be evaluated experimentally. (As an additional point of emphasis, the end-members of interest are strictly FCC because the precipitates targeted in study precipitate out of an FCC, γ -Ni matrix).

Table 3. Mobility data appended to .tdb (see Section II.II) [11], [12], [13]

Species i	$\phi_i^{\text{Ni,FCC}}$	$\phi_i^{\text{Fe,FCC}}$	$\phi_i^{\text{Nb,FCC}}$
Ni [11]	$D_0 = 3.44 \text{ cm}^2 \text{ s}^{-1}$	$D_0 = 1.12 \text{ cm}^2 \text{ s}^{-1}$	-
	$Q = 290.0 \text{ kJ mol}^{-1}$	$Q = 283.5 \text{ kJ mol}^{-1}$	-
Fe [11]	$D_0 = 9.20 \text{ cm}^2 \text{ s}^{-1}$	$D_0 = 0.72 \text{ cm}^2 \text{ s}^{-1}$	-
	$Q = 297.7 \text{ kJ mol}^{-1}$	$Q = 278.7 \text{ kJ mol}^{-1}$	-
Nb [12], [13]	$D_0 = 0.0104 \text{ cm}^2 \text{ s}^{-1}$	$D_0 = 0.75 \text{ cm}^2 \text{ s}^{-1}$	-
	$Q = 202.59 \text{ kJ mol}^{-1}$	$Q = 264 \text{ kJ mol}^{-1}$	-

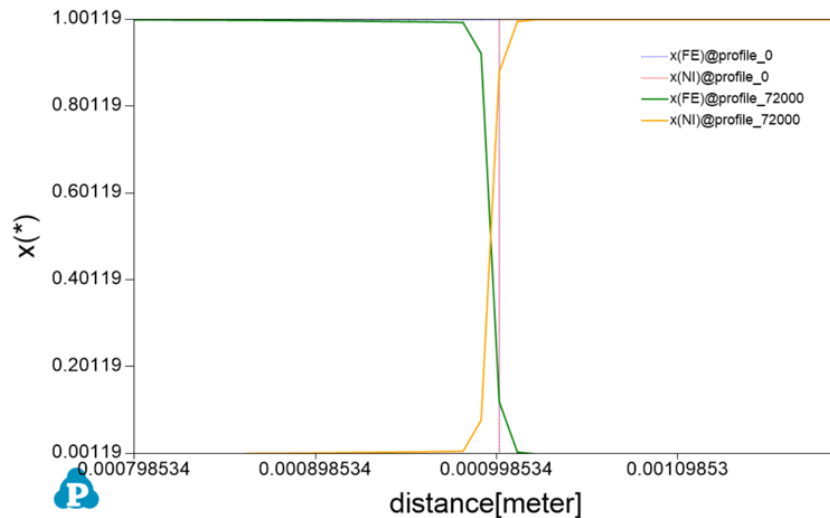


Figure 8. Simulated diffusion profile, Run 1, Table 1.

Figures 8-12 present the diffusion profiles calculated for the diffusion couples simulated according to Table 1. These diffusion profiles were calculated to test the mobility data appended to the .tdb developed for this study, Table 3. The concentration profiles are presented in the units of mole fraction of each component in the diffusion couple.

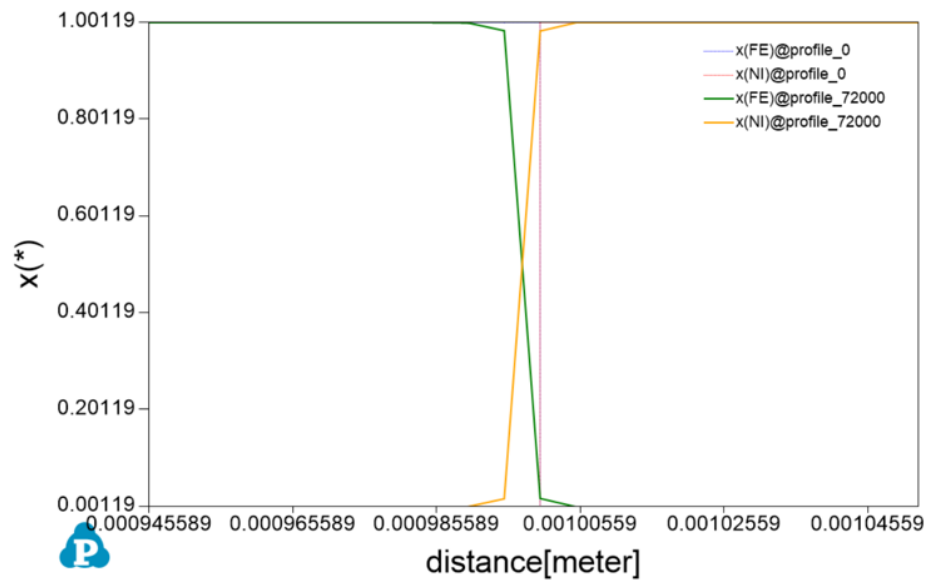


Figure 9. Simulated diffusion profile, Run 2, Table 1.

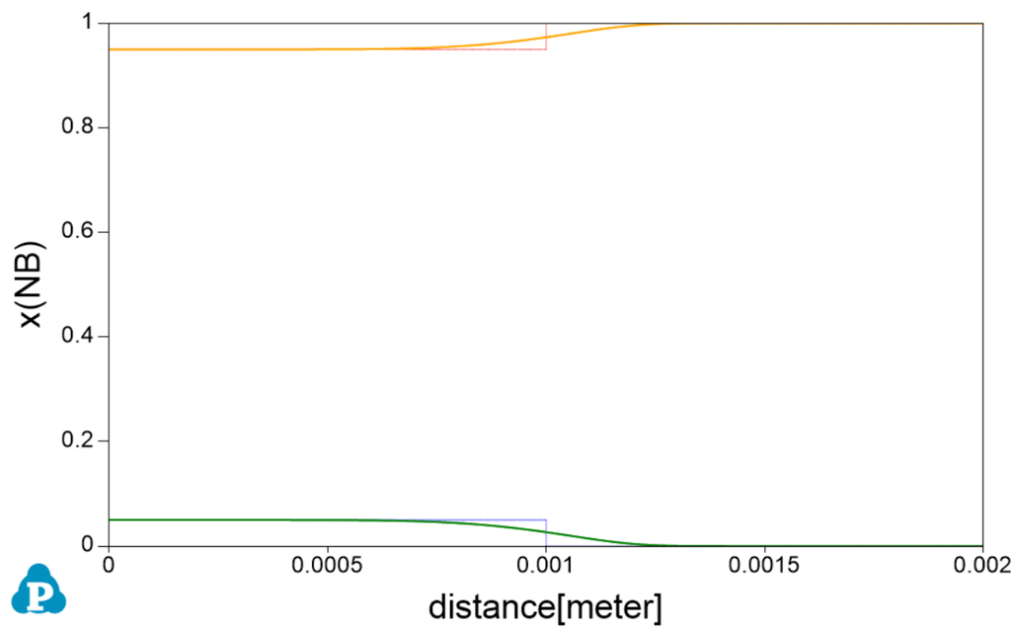


Figure 10. Simulated diffusion profile, Run 3, Table 1.

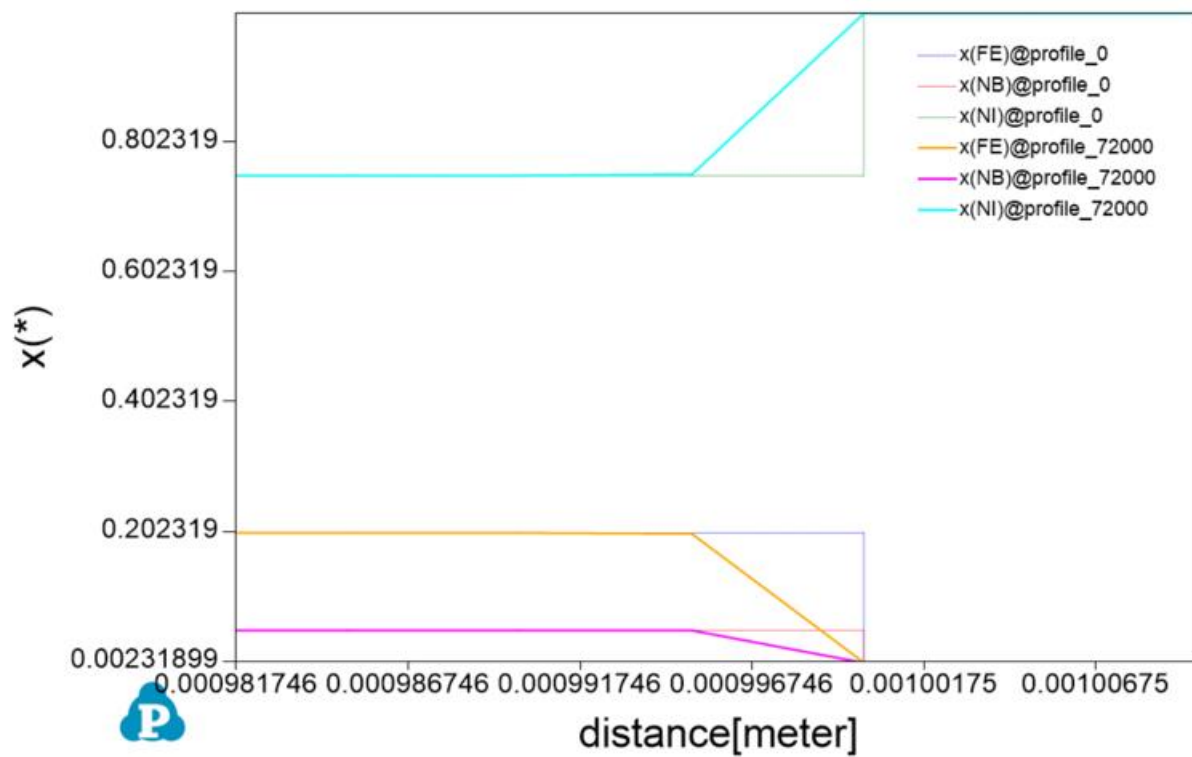


Figure 11. Simulated diffusion profile, Run 4, Table 1.

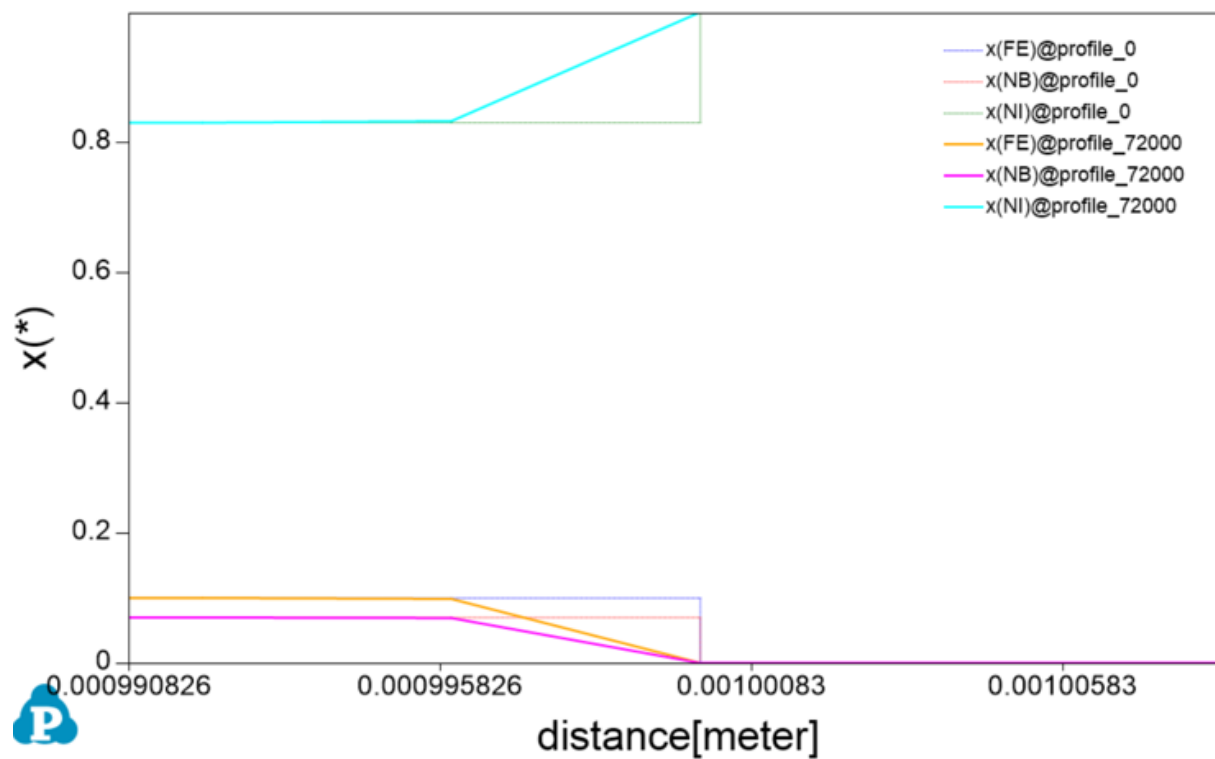


Figure 12. Simulated diffusion profile, Run 5, Table 1.

III.III Fe-Ni-Nb Precipitation Simulations

Table 4 presents the single-temperature precipitation simulations attempted to validate the functionality of the .kdb developed by this study. These cases were selected as well-established time-temperature conditions that lead to precipitation of γ'' and δ in commercial IN718, Figure 1. In all cases, Pandat failed to calculate any precipitation of the target secondary phases, pointing to inaccuracies with the .kdb developed or an inability to simulate the precipitation growth in IN718 using a ternary approximation of Fe-Ni-Nb. Figure 13 and Figure 14 illustrate representative volume fraction and particle size distribution curves for the precipitation simulations attempted by this study. The inability to precipitate secondary phases with the .kdb developed prevented the calculation of TTT and CCT diagrams for the Fe-Ni-Nb system.

Table 4. Attempted precipitation simulations using the developed .kdb

Run	C_L [at %]	T [C]	t [hr]	NSP, γ''	NSP, δ
1	20Fe-75Ni-5Nb	800	1	0.01	0.001
2	20Fe-75Ni-5Nb	950	10	0.01	0.001
3	22Fe-95Ni-3Nb	950	100	0.01	0.001
4	22Fe-95Ni-3Nb	950	100	0.1	0.01
5	22Fe-95Ni-3Nb	950	100	1	0.1

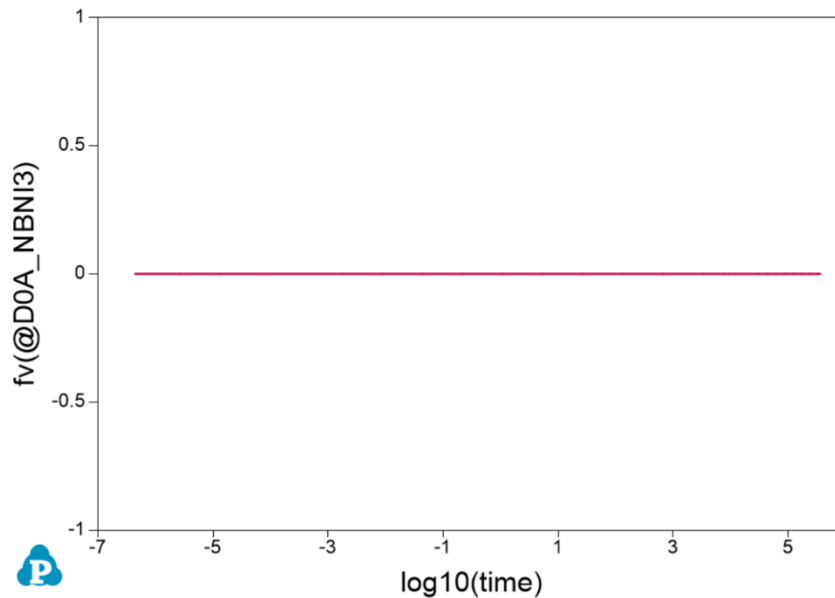


Figure 13. Representative volume fraction curve for all precipitation simulations presented in Table 4.

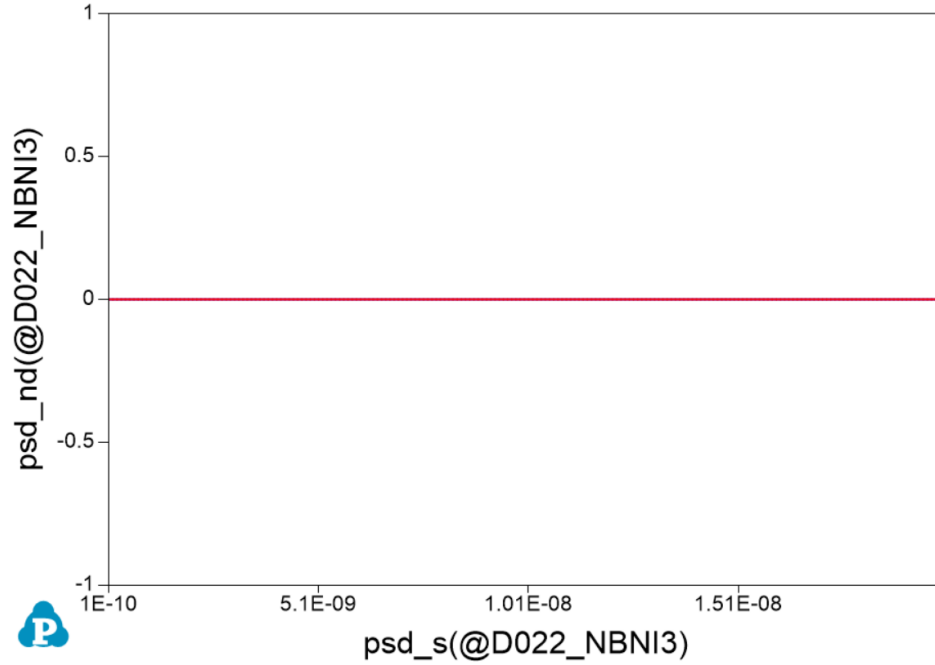


Figure 14. Representative particle size distribution for all precipitation simulations presented in Table 4.

IV. Discussion

The following section discusses the results presented in Section III. Through this discussion, the .tdb and .kdb files developed by this study are evaluated. Plausible thermodynamic and kinetic arguments are presented for why the .kdb developed by this study failed to predict precipitation of the expected secondary phases and why the Fe-Ni-Nb ternary approximation is unlikely an appropriate model of commercial IN718.

IV.I. Equilibrium Phase Diagrams and Evaluation of the Developed .tdb

The binary phase diagrams calculated using the .tdb developed by this study, Figures 2-4, reproduce the topology of the corresponding diagrams calculated by Mathon et al. [6] across the full temperature range relevant to the precipitation of the study's target secondary phases. The Fe-Ni phase diagram, Figure 2, captures the high-temperature γ -Ni (FCC_A1) phase field while the Ni-Nb phase diagram, Figure 3, reproduces the δ (D0A_NBNi3) phase field and its associated invariant reaction – important feature for this study. The Fe-Nb, Figure 4, phase

diagram does reproduce the published assessment; however, the intermetallic compounds predicted by this phase diagram are not relevant to this study as they were not considered in the kinetic database developed.

Like the binary phase diagrams, the isothermal ternary phase diagrams calculated at 723 K, 1273 K, and 1563 K (Figures 5-7 respectively) reproduced the principal features of the corresponding diagrams calculated by Mathon et al. [6]. The number, identity, and approximate composition of the equilibrium phase fields agree at all three temperatures, indicating a successful reproduction of the .tdb file. This confirms that the translation of the .tdb from Thermo-Calc to Pandat 2025 syntax, see Section II.I, was completed accurately, and the introduction of mobility data to the .tdb did not alter the underlying thermodynamic models. This makes the .tdb developed in this study a suitable foundation for future thermodynamic calculations of the Fe-Ni-Nb system.

IV.II. Diffusion Profiles and Evaluation of Mobility Data Appended to .tdb

All five diffusion profiles, Figures 8-12, illustrate the qualitative features expected of interdiffusion in a single phase with no phase transformations. The profiles are monotonic, featuring no oscillations, discontinuities, negative concentrations, or other artifacts of numerical instability. Although all profiles are monotonic, the degree to which the profiles resemble the expected sigmoidal curves differ due to grid spacing selections relative to the rate of diffusion in a given diffusion couple. For the cases where the concentration profile does not resemble a smooth sigmoidal curve, higher diffusion rates (via higher temperatures) or a finer grid spacing is required to achieve a smoothness comparable to the Ni-Nb diffusion profile.

The Ni-Nb diffusion couple, Figure 10, featured the smoothest, broadest concentration profile because of the high diffusivity of Nb in Ni, highlighting what the lower activation energy of Nb in Ni results in significantly higher diffusivities. Comparing Figure 8 and Figure 9 illustrates the effect of temperature on interdiffusion rates – the higher the temperature the wider the

interdiffusion zone because of higher diffusivities. This is consistent with the expected Arrhenius behavior of diffusion.

The two ternary diffusion couples simulated at 1473 K, Figure 11 and Figure 12, exhibit the features expected of multicomponent diffusion as each species develops its own interdiffusion zone depending on its mobility in the host matrix. The narrowness and linearity of the calculated ternary couple interdiffusion zones suggest that longer holding times and finer grid spacings are required for a true quantitative comparison to experimental data. However, the physically reasonable trends confirm that the mobility data appended to the .tdb, see Section II.II, was successfully implemented and produces reasonable kinetic behavior across the composition and temperature ranges of relevance to this study. The .tdb developed in this study is therefore a suitable thermodynamic *and* kinetic foundation for future diffusion simulations in a γ -Ni matrix for the Fe-Ni-Nb system.

IV.III Precipitation Simulations and Evaluation of the Developed .kdb

In contrast to the successful validation of the .tdb in Section IV.I and IV.II, the precipitation simulations in Section III.III failed to predict any nucleation of the target secondary phases, γ'' and δ , across all attempted conditions presented in Table 4. The calculated volume fraction of both phases remained zero regardless of variations in temperature of holding time, despite established TTT diagrams for commercial IN718 predicting a minimum of 1% volume fraction of each secondary phase for the conditions tested, Figure 1. This result is significant because the same .tdb that produced correct equilibrium phase fields and physically reasonable diffusion profiles was used by PanPrecipitation in these calculations, suggesting the kinetic parameters specified in the .kdb, Table 2, or approximating IN718 as Fe-Ni-Nb is the root cause of the lack of precipitation.

The first possibility to consider why precipitation cannot be modeled is that the .kdb is misconfigured. Comparing the kinetic parameters specified in the .kdb to published results, the interfacial energy assigned to γ'' in this study's .kdb (0.025 J m^{-2}) differs from the value reported by Devaux et al. for γ'' in commercial IN718 (0.095 J m^{-2}) [14]. However, this decrease in interfacial energy reduces the nucleation barrier, Equation 3, meaning precipitation should be easier to achieve. To test the effect of kinetic parameters on the calculated volume fraction of precipitates more directly, the nucleation-site parameter was increased by two orders of magnitude between Run 3 and Run 5 of the precipitation simulations carried out, Table 4. This drastic increase in the number of nucleation sites for both secondary phases produced no change in the calculated volume fraction of precipitates, ruling out availability of nucleation sites as a rate-limiting factor.

A second, plausible, explanation for why precipitates could not be formed is that the ternary approximation of IN718 tested in this study, Fe-Ni-Nb, does not provide sufficiently high thermodynamic driving forces or mobility for the precipitation of γ'' and δ . In commercial IN718, Cr occupies roughly 25 at. % of the γ -Ni matrix and significantly increases the activity coefficient of Nb in solution, increasing the supersaturation available to drive precipitation [15]. Reducing the composition of IN718 to Fe-Ni-Nb eliminates this driving force, slowing or eliminating the precipitation of secondary phases. Additionally, diffusion couple experiments of the Ti-Ni-Cr demonstrate a significant decrease in diffusivity when Cr is omitted from the system [16]. Since γ'' in commercial IN718 is not stoichiometrically pure Ni_3Nb , and contains approximately 71.1 at.% Ni, 17.5 at.% Nb, 4.3 at.% Ti, 1.0 at.% Al, 2.4 at.% Cr, 1.8 at.% Fe, and 1.9 at.% Mo [15], approximating IN718 as Fe-Ni-Nb and γ'' as only Ni_3Nb , is likely eliminating the complex interactions in chemical potential and atomic mobility driving nucleation in commercial IN718. For the purpose of predicting the kinetics of IN718's critical secondary phases, the alloy likely needs to be approximated as a more complex system, retaining elements such as Cr, Ti, and Al.

V. Conclusion

This study investigated the Fe-Ni-Nb system by developing a thermodynamic (.tdb) and kinetic (.kdb) database for the ternary system. The study calculated binary and ternary equilibrium phase diagrams to validate the thermodynamic foundation for the .tdb, diffusion profiles to validate the mobility models introduced to the .tdb, and precipitation simulations to assess the ability of the Fe-Ni-Nb .kdb to predict the precipitation of γ'' and δ in commercial IN718. This results of this study led to the following conclusions:

- The .tdb file reproduced by this study successfully calculates the equilibrium phase diagrams of the Fe-Ni-Nb system and can be used as a thermodynamic foundation for this ternary system.
- The mobility models appended to the .tdb calculate physically reasonable diffusion simulations results and can be used to model diffusion in a γ -Ni matrix.
- The .kdb file produced by this study fails to calculate the precipitation of secondary phases at temperature and time conditions expected to precipitate a minimum 1% volume fraction of secondary phases, according to established TTT diagrams of commercial IN718.
- Fe-Ni-Nb is not a suitable ternary approximation of commercial IN718 as it omits elements critical to the kinetic behavior of IN718, principally Cr, Ti, and Al. To carry out kinetic calculations of IN718, such as producing a TTT or CCT diagram, the alloy needs to be approximated as a more complex quaternary or higher-order system.

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